

SUMMARY

I am an assistant professor in the Computer Science department at Kunsan National University, South Korea. My research focuses on Human-Computer Interaction (HCI) and AR/VR, with an emphasis on developing frameworks and interactive techniques that make XR technologies more accessible and useful for everyday activities.

EDUCATION

2014-2022	Ph.D., Computer Science > Advisor: Prof. Daniel F. Keefe > Committee: Dr. Tobias Isenberg, Prof. Lana Yarosh, Prof. Evan Suma Rosenberg > Dissertation title: <i>Everyday Scientific Visualization: Making 3D Visualization Techniques Accessible for Day-To-Day Team-Science for Collaboration and Analysis</i> > Specializations: Data visualization, virtual reality	UNIVERSITY OF MINNESOTA – Minneapolis, MN
2012-2018	M.S., Computer Science > Specializations: Computer graphics, virtual reality	UNIVERSITY OF MINNESOTA – Minneapolis, MN
2008-2012	B.S., Computer Science > Specializations: Computer graphics, user interfaces	UNIVERSITY OF MINNESOTA – Minneapolis, MN

RELEVANT EXPERIENCE

2024-Present (1yr 11 mos)	Assistant Professor > Department of Computer Science and Information Engineering > Lead the Interaction Lab, developing frameworks and interactive techniques to make XR technologies more accessible and useful for everyday activities. Computer Graphics Game Engine Programming Capstone Design	KUNSAN NATIONAL UNIVERSITY – South Korea
2022-2024 (1 yr 11 mo)	Postdoctoral Researcher > Texas Advanced Computing Center. Advisor: Dr. Paul Navratil > Extended Intel's ray-tracing applications to support immersive VR experiences. > Extended the core rendering engine to support coherent visualization across large tiled displays. > Developed interaction techniques for gesture-based scene navigation and object manipulation. C++ CMake MPI Docker Intel OSPRay Microsoft Kinect Large Tiled Displays Python Open3D	UNIVERSITY OF TEXAS – Austin, TX
2019-2021 (2 yrs 6 mos)	Research Engineer > Korea Culture Technology Institute. Advisor: Prof. Moongu Jeon > Collaborated with external teams to design interactive installations for history museums. > Developed gesture-based interaction techniques for exploring archived cultural data. Unity C# Microsoft Kinect Large Format Displays	GWANGJU INSTITUTE OF SCIENCE AND TECHNOLOGY – Gwangju, S. Korea
2014-2019 (4 yrs 9 mos)	Research Assistant > Department of Computer Science and Engineering. Advisor: Prof. Daniel F. Keefe > Created interactive systems to assist scientists with analyzing and presenting their data > Collaborated on three multi-disciplinary projects involving teams at the U.S. National Forest Services, the Center for Spirituality and Healing, and the Medical Device Center Unity C# OptiTrack MS Kinect HMD VR CAVE VR Mobile VR Projection-based AR Processing 3D Printing	UNIVERSITY OF MINNESOTA – Minneapolis, MN

Summer 2018 (3 mos)	Research Intern INRIA – Saclay, France ➤ Aviz. Advisor: Dr. Tobias Isenberg ➤ Investigated ways of leveraging storytelling and lightweight communication for science collaboration ➤ Developed frameworks for creating data stories and collaborating around exchanged stories in different device settings, e.g., in browsers, phones, and desktop settings Unity C# PHP MySQL JavaScript CSS HTML
2011-2014 (3 yrs 3 mos)	Programmer UNIVERSITY OF MINNESOTA – Minneapolis, MN ➤ Center for Magnetic Resonance Research. Advisor: Prof. Greg Metzger ➤ Worked with pathologists to develop a Photoshop-like Java application for assembling scanned tissue images into a complete organ and annotating cancer boundaries for further data analysis ➤ Integrated Java3D to view and interact with drawn cancer boundaries in 3D and implemented corresponding interaction functionalities Java Java3D

PUBLICATIONS

- 2026 **J. W. Nam**, V. Chheang, and H. Miao, “Spatialslides: Integrating presentation software with extended reality for contextualized 3D exploration,” in *2026 ACM symposium on virtual reality software and technology (VRST)*, 2026, to appear
[▶ YouTube](#)
- Poster - J. Jang and **J. W. Nam**, “Understanding task resumption strategies after physical interruptions in 360 VR videos,” in *2026 IEEE international symposium on mixed and augmented reality adjunct (ISMAR-Adjunct)*, 2026, to appear
[▶ YouTube](#)
- Poster - A. Chandanshive and **J. W. Nam**, “Mapping the design space of XR interruption delivery: Initial framework,” in *2026 IEEE international symposium on mixed and augmented reality adjunct (ISMAR-Adjunct)*, 2026, to appear
[▶ YouTube](#)
- 2024 **J. W. Nam**, T. Isenberg, and D. F. Keefe, “V-Mail: 3D-enabled correspondence about spatial data on (almost) all your devices,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 4, pp. 1853–1867, 2024. doi: 10.1109/TVCG.2022.3229017
[▶ YouTube](#) [▶ Presentation IEEE VIS](#)
- 2023 **J. W. Nam**, G. D. Abram, F. Samsel, and P. A. Navrátil, “Immersive OSPRay: Enabling VR experiences with OSPRay,” in *2023 ACM conference on practice and experience in advanced research computing (PEARC)*, 2023, p. 226–230. doi: 10.1145/3569951.3597579
- 2022 D. F. Keefe, B. Herman, **J. W. Nam**, D. T. Orban, and S. Johnson, “Hybrid data constructs: Interacting with biomedical data in augmented spaces,” in *Making Data: The Creative Practice of Materialising Digital Information*. London: Bloomsbury, 2022, ch. 11, pp. 169–182. doi: 10.5040/9781350133266.ch-011
- 2019 **J. W. Nam** et al., “Worlds-in-wedges: Combining worlds-in-miniature and portals to support comparative immersive visualization of forestry data,” in *2019 IEEE conference on virtual reality and 3D user interfaces (VR)*, 2019, pp. 747–755. doi: 10.1109/VR.2019.8797871
[▶ YouTube](#) [▶ Presentation IEEE VR](#)
- E. L. et al., “Signature maps for automatic identification of prostate cancer from colorimetric analysis of h&e-and ihc-stained histopathological specimens,” *Nature Scientific Reports*, vol. 9, no. 6992, 2019. doi: 10.1038/s41598-019-43486-y
- Poster - **J. W. Nam**, C. H. Perry, B. T. Wilson, and D. F. Keefe, “Linked view visualization using clipboard-style mobile VR: Application to communicating forestry data,” *IEEE VIS Posters*, 2019
[▶ YouTube](#) [🌟 SciVis Best Poster Award](#)

Poster - N. Park, Y. Hong, H. Park, **J. W. Nam**, K. Kim, J. Pyo, K. Gil, and K. Lee, “Effects of age and motivation for visiting on AR museum experiences,” ACM VRST Posters, 2019. doi: 10.1145/3359996.3364711

2017 **J. W. Nam** and D. F. Keefe, “Spatial correlation: An interactive display of virtual gesture sculpture,” *Leonardo*, vol. 50, no. 1, pp. 94–95, 2017. doi: 10.1162/LEON_a_01226

 [YouTube](#)

2016 H. Farooq, J. Xu, **J. W. Nam**, D. F. Keefe, E. Yacoub, T. Georgiou, and C. Lenglet, “Microstructure imaging of crossing (MIX) white matter fibers from diffusion MRI,” *Nature Scientific Reports*, vol. 6, no. 38927, 2016. doi: 10.1038/srep38927

G. J. M. et al., “Detection of prostate cancer: quantitative multiparametric MR imaging models developed using registered correlative histopathology,” *Radiology*, vol. 279, no. 3, pp. 805–816, 2016. doi: 10.1148/radiol.2015151089

RESEARCH GRANTS

2026–2028 **Young Scientist Grants (Type A)**, National Research Foundation of Korea (NRF). Principal Investigator. Three-year research grant (KRW 80 million/year in direct funding) supporting research on seamless VR task resumption through recovery of prior interaction states.

AWARDS

October 2019 **SciVis Best Poster Award**, “Linked View Visualization Using Clipboard-Style Mobile VR: Application to Communicating Forestry Data,” Poster track at IEEE VIS 2019. Vancouver, British Columbia, Canada.

PUBLIC EXHIBITIONS

December 2021 Developer, “The Road of Hyecho.” Interactive installation at Gwangju Cultural Foundation. S. Korea.

 [News](#)

Unity

C#

Microsoft Kinect

Projection Wall

December 2020 Developer, “The Road of Ramayana.” Interactive installation at Asia Culture Center. Gwangju, S. Korea.

 [YouTube](#)

 [News](#)

 [News](#)

Unity

C#

Microsoft Kinect

Large Format Display

November 2014 Developer, “Spatial Correlation: An Interactive Display of Virtual Gesture Sculpture.” Interactive installation at IEEE VIS 2014 Arts Program. Paris, France.

 [YouTube](#)

 [Publication](#)

Processing

Java

GLSL

Microsoft Kinect V1

PRESENTATIONS

October 2024 Presenting author, “V-Mail: 3D-Enabled Correspondence About Spatial Data on (Almost) All Your Devices ” Paper presentation at IEEE VIS 2024 (virtual).

July 2023 Presenting author, “Immersive OSPRay: Enabling VR experiences with OSPRay.” Paper presentation at ACM PEARC 2023, Portland, Oregon, USA.

March 2019 Presenting author, “Worlds-in-Wedges: Combining Worlds-in-Miniature and Portals to Support Comparative Immersive Visualization of Forestry Data.” Paper presentation at IEEE VR 2019, Osaka, Japan.

PROFESSIONAL SERVICES

Reviewer TVCG (2023, 2024, 2026), ISMAR (2024), VR (2022), DESRIST (2018), Leonardo (2016)

Panelist The Future of AI that Communicates with Humans, Hwangryong Humanities Talk Concert, 2025.10.31